

BRAC^T's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI

Semester: 5

Academic Year: 2026-27

Division: E

Batch: 1

Practical No: 1

Roll no: 52

PRN No: 12410371

Name of Student: Vaidik Samarth

Subject Name: Deep Learning

Problem Statement: Perform data preprocessing, normalization, train-test splitting, and data visualization on a sample dataset.

Code and Output:



The screenshot shows a Jupyter Notebook interface with a title bar and a toolbar. The notebook is titled "Assignment 1 : Perform data preprocessing, normalization, train-test splitting, and data visualization on a sample dataset." The code is written in Python and uses TensorFlow and Matplotlib. The output shows the shapes of the training and testing images and labels, and a visualization of the first training image in grayscale.

```
import tensorflow as tf
import matplotlib.pyplot as plt

(train_images, train_labels), (test_images, test_labels) = tf.keras.datasets.mnist.load_data()

train_images = train_images / 255.0
test_images = test_images / 255.0

print(train_images.shape)
print(test_images.shape)
print(train_labels)

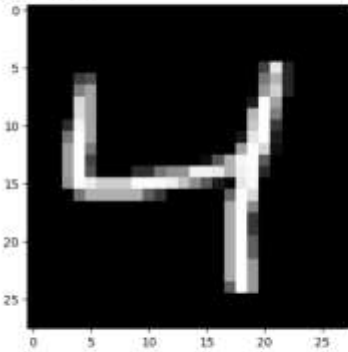
# Display first image
plt.imshow(train_images[1], cmap='gray')
plt.show()

(60000, 28, 28)
(10000, 28, 28)
```

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```
plt.imshow(train_images[2], cmap= 'gray')
plt.show()
```

```
(60000, 28, 28)
(10000, 28, 28)
[5 0 4 ... 5 0 8]
```



```
my_model = tf.keras.models.Sequential()
```

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```
[10] my_model = tf.keras.models.Sequential()
      my_model.add(tf.keras.layers.Flatten(input_shape=(28, 28)))
      my_model.add(tf.keras.layers.Dense(128, activation='relu'))
      my_model.add(tf.keras.layers.Dense(10, activation='softmax'))

      # Compiling the model
      my_model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

      # Training the model
      my_model.fit(train_images, train_labels, epochs=3)
```

```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/reshaping/flatten.py:37: UserWarning: Do not pass an 'input_shape' / 'input_dim' argument to a layer. when using Sequential().
  super().__init__(**kwargs)
Epoch 1/3
1875/1875 — 7s 3ms/step - accuracy: 0.8254 - loss: 0.2574
Epoch 2/3
1875/1875 — 5s 3ms/step - accuracy: 0.9060 - loss: 0.1145
Epoch 3/3
1875/1875 — 6s 3ms/step - accuracy: 0.9761 - loss: 0.0752
keras.src.callbacks.history.History at 0x7c9b1984e660
```

```

[11] val_loss, val_acc = my_model.evaluate(test_images, test_labels)
      print('Test accuracy: ', val_acc)
```

```

318/318 — 1s 3ms/step - accuracy: 0.9702 - loss: 0.0665
Test accuracy: 0.970200021934589
```

```

[12] my_model.save('my_mnist_model.keras')
```

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